

July 22, 2026 – PCB Design

Over the past week, I completed the design of my first custom printed circuit board (PCB) using **KiCad**. The board is a **two-layer PCB** designed for my RC Crawler Telemetry System, which integrates an ESP32 microcontroller with multiple sensors, including a BNO055 IMU, Hall effect sensor, DS18B20 temperature sensor, and a battery voltage monitoring circuit. I completed the schematic, component placement, PCB routing, ground planes, and resolved all design rule check (DRC) errors to produce a manufacturable design.

After completing the PCB layout, I generated the Gerber and drill files, verified the design using JLCPCB's online Gerber viewer, and submitted the board for manufacturing. I also committed the completed PCB design to my GitHub repository to document this milestone in the project.

My next steps are to assemble the PCB once it arrives by soldering the components, connecting the buck converter and sensors, and testing the hardware with my existing ESP32 firmware and Flutter mobile dashboard to verify that all telemetry data is transmitted correctly.

